

Bachelor in Computer Applications

Course Title: **Operating System**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Why Optimal Page Replacement is best but not practically feasible page replacement algorithm? Calculate the number of page faults for Optimal, LRU and FIFO replacement algorithm for the reference string: 1, 3, 4, 2, 3, 5, 4, 3, 1, 2, 4, 6, 3, 2, 1, 4, 2 using 3 page frames.
2. Find Average waiting time and turn-around time for following example using FIFO, SRTF and Round Robin scheduling algorithm. Assume quantum as 4 ms.

Process	id	Arrival	time	Burst
P108	P215	P3110	P4213	P5217
3. Why virtual memory technique is used in the computer system? What is logical address? Explain the process of conversion of logical address to physical address using TLB.
4. Suppose that a disk drive has the cylinder numbered, 0 to 199 is currently serving a request at cylinder 143. The request queue is kept in the FIFO order 25, 17, 119, 197, 194, 15, 182, 115, and 183. What is the total head movement needed to satisfy these request for the following disk scheduling algorithm. a) FCFS b) SSTF
5. If a 2gb disk has 4-KB block size, calculate the size of the file allocation table if each entry of the table is 4 bytes.
6. What is process? Explain 3 state process models with suitable diagram.
7. Write down the basic difference between coalescing and compaction with diagram.
8. What is memory mapped I/O? Explain about device independent I/O software.
9. Differentiate between deadlock and starvation? Discuss the process of detecting deadlocks when there are multiple resources of each type.
10. Differentiate between process and a program. What is PCB?
11. Write down the Solving technique of the producer consumer problem with Message passing?
12. What is system call? Explain the system call process in detail.